

# Dr. Marcel Brosch

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## EDUCATION

2022 – now

October

### **Postdoctoral Researcher**

Tool development addressing current challenges in neuroscience & medicine, focus on one-photon imaging with miniature microscopes

Asst. Prof. Dr. D. Aharoni

UCLA, Department of Neurology, Los Angeles, United States

2016 - 2022

April – Mai

### **Doctor of Natural Science** (summa cum laude)

Development and application of neuroscience tools combining electrophysiology and optophysiology: a transparent  $\mu$ ECoG array and optrode microdrive

Prof. Dr. F. Ohl, Dr. M. Lippert

Leibniz-Institute for Neurobiology, Magdeburg, Germany

2013 - 2016

### **Master of Science, Integrative Neuroscience**

Functional auditory cortico-striatal coupling during two-way active avoidance reinforcement paradigms

Prof. Dr. F. Ohl, Dr. M. Woldeit

Leibniz-Institute for Neurobiology, Magdeburg, Germany

2010 - 2013

### **Bachelor of Science, Biology**

Optimization of the Barnes maze protocol for detection of differences in spatial learning of cyclin D2 knockout mice and wild type mice.

Prof. Dr. K. Lehmann, Dr. A. Urbach

Clinic for Neurology, University Hospital Jena, Germany

Languages:

native speaker German, fluent in English

## PUBLICATIONS

**Brosch M**, Vlasenko A, Ohl FW, Lippert MT (2021) TetrODrive: An open-source microdrive for combined electrophysiology and optophysiology. *J Neural Eng* . [18:046030](#)

**Brosch M**, Deckert M, Rathi S, Takagaki K, Weidner T, Ohl FW, Schmidt B, Lippert MT (2020) An optically transparent multi-electrode array for combined electrophysiology and optophysiology at the mesoscopic scale. *J Neural Eng*. [17\(4\):046014](#)

Coors A, **Brosch M**, Kahl E, Khalil R, Michels B, Laub A, Franke K, Gerber B, Fendt M (2019) Rhodiola rosea root extract has antipsychotic-like effects in rodent models of sensorimotor gating. *J Ethnopharmacol*. [235:320-328](#)

Xia Z, Arias-Gil G, Deckert M, Vollmer M, Curran A, Herrera-Molina R, **Brosch M**, Krug K, Schmidt B, Ohl FW, Lippert ML, Takagaki K (2019) Electrochemical roughening and carbon nanotube coating of tetrodes for chronic single-unit recording *bioRxiv* [738245](#)

Fendt M, **Brosch M**, Wernecke KEA, Willadsen M, Wöhr M (2018) Predator odour but not TMT induces 22-kHz ultrasonic vocalizations in rats that lead to defensive behaviours in conspecifics upon replay. *Sci Rep*. [8\(1\):10041](#)

Tegtmeier J, **Brosch M**, Janitzky K, Heinze H-J, Ohl FW, Lippert MT (2018) CAVE: An open-source tool for combined analysis of head-mounted calcium imaging and behavior in MATLAB. *Front Neurosci*. [12:958](#)

Schulz AL, Woldeit ML, **Brosch M**, Ohl FW (2017) Neurobiological fundamentals of strategy change — A core competence of a companion system. In: International Conference on Companion Technology: *IEEE* [8287074](#)

## DATA SCIENCE

2022 January – March

### **Data Analytics Graduate** (full-time bootcamp)

- Python Data analytics & MySQL
- Build, evaluate & deploy machine learning models (Scikit-Learn)
- Inferential statistics, use of API & Web scraping, Data visualization

Ironhack, Remote learning Campus Amsterdam

Lead teacher: Jan Molendjik

## ADDITIONAL RESEARCH EXPERIENCE

2015 (March) - 2016 (April)

### **Research Assistant, Collaborative Research Center**

- Rodent behavioral auditory discrimination task and *in vivo* electrophysiological
- Custom written MATLAB analysis scripts

Dr. Schulz and Dr. Woldeit

Leibniz-Institute for Neurobiology, Magdeburg

2015 (October)

### **Field work assistant, Leibniz-Institute for Zoo and Wildlife Research**

- collecting wild mice in rural farmlands, preparation for genetic analysis

Prof. Dr. Emanuel Heitlinger

Humboldt-University, Berlin

2014 (April) – 2015 (March)

### **Research Assistant, Neuropharmacology of Emotional Systems**

- ultrasonic vocalization in rodents during relief- and fear conditioning paradigms
- Pharmacological effect of plant extracts on prepulse inhibition

Prof. Dr. Fendt

Institute for Pharmacology and Toxicology, Magdeburg

2014 (September)

### **Research Assistant, Neurotrophins in Emotional Learning**

- stereotactic pharmacological treatments in rodents and fear conditioning

Dr. Endres

Institute of Physiology, Magdeburg

2012 (August-October)

### **Research Assistant, Neurobiological Principles of Pain Development**

- Primary cell culture work on neurobiological basics of pain development and the neuronal control system of inflammation

Prof. Schaible and Dr. König

Institute for Neurophysiology, Jena

2011 (August)

### **Research Assistant, Plant biodiversity and ecosystem functioning**

- Field work: plant diversity studies

Jena, German Center for Integrative Biodiversity Research

## INVITED INTERNATIONAL CONFERENCE TALKS

- 2019                      Breaking news symposium: A flexible and transparent electrode array for closed-loop optogenetic stimulation  
13<sup>th</sup> German Neuroscience Meeting,  
March 20.- 23. Göttingen, Germany
- 2016                      Young physiologists symposium: A systemic view of learning induced brain network changes: illustrating memory formation and neuronal circuit plasticity  
95<sup>th</sup> Annual Meeting of the German Physiology Society  
March 3. – 5., Lübeck, Germany
- 2015                      Best talk award: Auditory cortico-striatal coupling during different two-way active avoidance task paradigms  
4<sup>th</sup> Symposium of the Young Physiologists  
September 24. - 25., Leipzig, Germany

## CONFERENCE PRESENTATIONS

- Society for Neuroscience (SFN) Conference                      Temporal evolution of reward prediction error signals during reward and punishment learning  
Poster presentation, 13.-17. November 2019, USA, Chicago
- Federation of European Neuroscience Society (FENS) Conference                      A new flexible and transparent multichannel surface-electrode  
Poster presentation, 07.-11. July 2018, Germany, Berlin
- International Conference on Auditory Cortex                      Novel flexible and transparent  $\mu$ ECoG electrode arrays for the measurement of mesoscopic activity in auditory cortex  
Poster presentation, 10.-15. September 2017, Canada, Banff

## SCIENCE POLICY AND SCIENCE COMMUNICATION

- 2017-2019                      **Ph.D. student representative**  
Leibniz-Institute for Neurobiology, Magdeburg  
Graduate student coordinator and organizer of graduate student conferences,  
Member of the nationwide graduate student Leibniz network
- 2019                      **Founder of [Simpleneuro.org](http://Simpleneuro.org)**  
A blog of young scientist for the general public and other young scientists  
Strong interest and passion in science communication

## REFERENCES

Prof. Dr. F. Ohl  
Leibniz Institute for Neurobiology  
Systemsphysiology Department

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Prof. Dr. M. Fendt  
Otto-von-Guericke University  
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